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Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction

Completed Research Paper

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Abstract

Cultural crowdfunding (CCF) enables creators to showcase artistic and creative work while seeking public support, but existing success prediction models often neglect the cultural attributes inherent in such projects. To address this gap, we propose the Cultural Value-based Prediction Model (CV-PM), grounded in Throsby's cultural value framework. CV-PM systematically extracts and integrates six cultural values from multimodal project content. Beyond feature modeling, our model also uncovers how cultural values interact and contribute to outcomes, thereby operationalizing the cultural mechanisms. In evaluations on 14,539 Kickstarter projects, CV-PM outperforms traditional and multimodal baselines. Additional analyses validate the contribution of each cultural value and reveal domain-specific patterns. By bridging cultural theory with deep learning, this study offers a theoretically grounded and interpretable framework for CCF success prediction. It also provides practical insights for creators and platforms, enabling cultural design and strategy optimization in creative industries.

Keywords: Crowdfunding success prediction, theory-guided deep learning, cultural crowdfunding, cultural values framework, multimodal data

Introduction

Cultural crowdfunding (CCF) is a comprehensive process whereby creators utilize crowdfunding as a sales channel for their artistic and creative work to increase their professional reputation, market visibility, and recognition, and to obtain supplemental funding (Kappel, 2008). In recent years, the global crowdfunding market has witnessed significant expansion, growing from USD 17.72 billion in 2024 to USD 20.46 billion in 2025, reflecting a compound annual growth rate (CAGR) of 15.5%[1]. This rapid growth has attracted a growing number of creators and backers from the cultural and creative industries (CCIs[2]), establishing CCF as a salient topic in both practice and academic discourse.

[1] The Business Research Company, "Crowdfunding Global Market Report 2025".

[2] According to the 'Creative Europe Programme' established by Article 2 (Point 1) of Regulation (EU) n. 1295/2013 of the European Parliament and of the Council of 11 December 2013, the cultural and creative industries can be defined as: "all sectors whose activities are based on cultural values and/or artistic and other creative expressions, whether those activities are market- or non-market-oriented, whatever the type of structure that carries them out, and irrespective of how that structure is financed."

Reward-based crowdfunding is particularly compatible with CCF, as it enables the presentation of rich creative content while fostering community engagement through compelling reward structures. For instance, Kickstarter[3], one of the world's largest reward-based platforms, hosts projects primarily within CCF-related categories such as film, music, publishing, design, and art. However, Kickstarter reported a global project success rate of only 42.19%, which means that the majority of projects fail, resulting in considerable losses of time and resources for platforms, creators, and backers alike[4]. Consequently, reliable pre-launch success prediction is essential, as it enables early evaluation of project viability before launch, allowing creators to refine project design and platforms and backers to allocate resources more effectively.

To support crowdfunding success prediction, prior research has mainly concentrated on four categories of success determinants. Firstly, project-related determinants capture the intrinsic attributes of a project, including funding target (Liu et al., 2022), project duration (Huo et al., 2024), reward structure (Shi, 2018), project category (McSweeney et al., 2022), and media richness (Yeh et al., 2019). Secondly, creator-related determinants reflect characteristics of the project initiator—such as gender (Greenberg & Mollick, 2017), race (Koh et al., 2023), hedonic value (Zheng et al., 2017), and the extent of self-funding efforts (L. Zhao et al., 2022). Thirdly, backer-related determinants focus on their behavioral patterns, including altruistic tendencies (Cox et al., 2018), engagement in positive interactions (Wei et al., 2024), and overall motivational profiles (Bretschneider & Leimeister, 2017). Finally, platform-related determinants—such as the number of concurrent projects (Chan et al., 2021), platform type (Solodoha & Blaywais, 2023), and the age of the platform (Shneor et al., 2020)—have also been found to affect project outcomes.

While prior studies have explored a broad range of determinants related to crowdfunding success, they have paid limited attention to the intrinsic nature of cultural products. This gap is particularly salient in the context of CCF, where projects often depart from conventional goods in both purpose and presentation. As Throsby (2001) contends in *Economics and Culture* (Throsby, 2001), cultural products are inherently infused with creativity, symbolic meaning, identity, and heritage—attributes that render their evaluation uniquely complex and context-dependent. Building on Throsby's cultural value framework, cultural value has been conceptualized as a multidimensional construct comprising six core dimensions: aesthetic value, spiritual value, social value, historical value, symbolic value, and authenticity value (Throsby, 2001). Recent studies further support the importance of cultural value in CCF. For example, Dematos Guimarães and Maehle (2023) and Rykkja et al. (2020) emphasized that the cultural significance of a project is a potential driver of success (Dematos Guimarães & Maehle, 2023; Rykkja et al., 2020). Despite such insights, existing research remains fragmented—frequently relying on qualitative descriptions or focusing on isolated cultural values—without offering a comprehensive framework to quantify and operationalize cultural values in predictive models. Motivated by this gap, our study seeks to address the following research questions:

RQ1: How can the multimodal content of crowdfunding projects be systematically mapped to cultural values grounded in Throsby's cultural value framework, ensuring both coverage and interpretability?

RQ2: Through what mechanisms do these cultural values influence crowdfunding success?

To address these research questions, we confront two key challenges. First, cultural features are often implicitly embedded within CCF content, making them difficult to conceptualize and prone to noise and redundancy. Second, understanding the influence of cultural values on crowdfunding success requires not only analyzing each value dimension individually but also accounting for their interdependencies in a coherent and integrated manner. In order to overcome these challenges, we developed the Cultural Value-based Prediction Model (CV-PM) for CCF, comprising three interrelated modules: (i) First, the Cultural Value Modeling (CVM) maps multimodal project content onto six cultural values by aligning with established theoretical frameworks. (ii) Second, the Deep Cultural Embedding Refinement (DCER) module compresses these features into high-level representations using convolutional neural networks (CNNs). (iii) Third, the Comprehensive Prediction Module (CPM) module employs a Cultural Value Fusion Attention Mechanism (CVFAM) to assign weights to different cultural values and aggregate them for the final prediction.

[3] Kickstarter (https://www.kickstarter.com/) is a reward-based crowdfunding platform with All-or-Nothing funding mechanism, which maintains a global crowdfunding platform focused on cultural and creativity. For more statistical details, visit its official website: Kickstarter Stats — Kickstarter.

[4] Stats sourced from the official website as of April 2025: Kickstarter Stats — Kickstarter

This study offers several contributions to the literature on CCF and cultural analytics:

(i) From a theoretical standpoint, it introduced a cultural lens into the modeling of crowdfunding success by quantitatively examining the role of cultural values across project types.

(ii) Methodologically, CV-PM provides a systematic approach to extracting and integrating cultural features from metadata, textual descriptions, and visual content, while preserving theoretical interpretability through representation learning.

(iii) Empirically, evaluation on a dataset of 14,539 Kickstarter projects showed that CV-PM outperformed baseline methods, achieving an accuracy of 90.86%, an AUC of 95.14%, and an F1-score of 93.08%.

(iv) Practically, the model enabled quantification of the relative importance of cultural values within specific categories, offering creators actionable guidance on how to enhance success probabilities by emphasizing culturally salient aspects of their projects.

Related Literature

This section reviews the literature relevant to our study, focusing on methods for predicting crowdfunding success and the theoretical foundations and applications of Throsby's cultural value framework.

Prediction Methods for Crowdfunding Success

A wide range of methods have been employed to predict crowdfunding success, evolving from traditional machine learning-based prediction methods to more advanced single-modal and multi-modal deep learning-based prediction methods that incorporate diverse forms of project content.

Traditional machine learning-based prediction methods typically rely on handcrafted features derived from structured data and apply models such as logistic regression, random forests, and XGBoost. For instance, Li et al. (2016) applied logistic regression to evaluate how funding goals and project durations influence success rates (Li et al., 2016). Greenberg et al. (2013) utilized random forest models to assess the impact of updates, comments, and reward levels on funding outcomes (Greenberg et al., 2013). More recently, Porro et al. (2024) employed XGBoost to predict crowdfunding performance based on a variety of metadata features, including investor behavior and investment patterns (Porro et al., 2024). Although effective in certain contexts, these methods often struggled to capture complex, nonlinear relationships among features, limiting their ability to model complex crowdfunding projects.

To address these limitations, researchers have turned to single-modal deep learning-based prediction methods, which typically apply neural architectures to a single modality—such as metadata, text, or visual contents. Zhong et al. (2022) proposed a model using metadata features, including funding goal, reward count, and the number of historical projects, along with network embeddings derived from project connections, significantly improved predictive accuracy (Zhong et al., 2022). Du et al. (2021) introduced a backer-centric approach that measures persuasibility by comparing backers' language preferences with project descriptions, using CNN and SVM for prediction (Du et al., 2021). Blanchard et al. (2023) showed that visual features—such as brightness, contrast, number of faces, and image recognizability—along with media quantity, are strong predictors of success (Blanchard et al., 2023). Although single-modality deep learning-based prediction methods advanced the field by enabling nonlinear modeling, their limited scope reduced their capacity to capture the full complexity and richness of crowdfunding presentations.

In response, multi-modal deep learning-based prediction methods have gained traction. These methods jointly consider metadata, text, and visual content to better approximate the diverse signals embedded in project presentations. Cheng et al. (2019) developed a joint model combining metadata, text, and visual features, showing that visual contents significantly improved performance, particularly when text was limited (Cheng et al., 2019). Tang et al. (2023) introduced a cross-attention mechanism aligning video frames with project narratives, enabling the identification of influential multimodal elements (Tang et al., 2023). Cai et al. (2024) advanced this line by modeling feature interactions through a dynamic graph convolutional network that jointly processes text and visual contents (Cai et al., 2024).

Despite these methodological advancements, prior work has largely overlooked the cultural signals embedded in project content. This is particularly notable in CCF, where creators frequently express

meaning through narrative, symbolism, and visual form (Kappel, 2008). The existing studies have suggested that the cultural relevance of CCF can meaningfully shape its chances of success (Rykkja et al., 2020). Nonetheless, the literature to date remains fragmented, often relying on descriptive analysis or treating culture in a piecemeal fashion, without a unified framework for its systematic quantification and integration into predictive modeling.

Cultural Value Frameworks for Crowdfunding

Most research on cultural crowdfunding has been grounded in economic perspectives, while studies explicitly adopting cultural theories remain relatively scarce. Two theoretical approaches have received particular attention: Hofstede's national cultural framework and Throsby's cultural value framework.

Hofstede (1984) conceptualizes culture along six dimensions, i.e., power distance, individualism versus collectivism, masculinity versus femininity, uncertainty avoidance, long- versus short-term orientation, and indulgence versus restraint (Hofstede, 1984), which capture differences in hierarchy, autonomy, and tolerance for uncertainty. In crowdfunding, Cicchiello et al. (2023) show that these macro-level cultural traits shape adoption in the cultural and creative industries (Cicchiello et al., 2023). Yet, because Hofstede's framework operates at the national level and is derived from large-scale surveys, it cannot be directly mapped onto project-level multimodal content, limiting its interpretability for predicting project success.

To address this gap, we draw on Throsby's cultural value framework. Throsby (2001) argued that cultural products should be assessed not only by economic returns but also by intrinsic cultural significance, proposing six dimensions, i.e., aesthetic, spiritual, social, historical, symbolic, and authenticity, as a structured yet flexible lens. He later reinforced its relevance in policy debates, stressing the difficulty of quantifying cultural value without reducing its complexity (Throsby, 2001). This framework thus provides a theoretical foundation for understanding how cultural content conveys meaning, invokes collective memory, and sustains social cohesion. Recent studies further illustrate how these six values manifest in digital and participatory environments. In CCF, aesthetic and symbolic values often appear through narrative design, visuals, and branding that shape emotional resonance and perceived quality (Demattos Guimarães & Maehle, 2023). Social and spiritual values surface in projects emphasizing participation, community ties, or emotional uplift, often examined via participant feedback (Sacco, 2011). Historical and authenticity values are salient in heritage and creative industry projects, where tradition and credibility foster trust (Jones, 2010; Mason, 2002). Although still underexplored in crowdfunding, existing evidence suggests that articulating cultural value can shape backer perceptions and influence funding outcomes, making this framework a promising lens for understanding CCF success.

Cultural Value-based Prediction Model

Overview of CV-PM

To address the two research questions—mapping multimodal content to cultural values (RQ1) and understanding how these values influence success (RQ2)—we propose the CV-PM, a framework specifically tailored for the CCF context. As shown in Figure 1, CV-PM consists of three core modules that systematically operationalize cultural values and integrate them into the prediction process.

First, the CVM maps project content from metadata, text, and visual content onto a set of cultural values derived from Throsby's cultural value framework, producing interpretable cultural features. Next, the DCER applies CNNs to compress these features into deep cultural embeddings that preserve semantic meaning while reducing noise and redundancy. Finally, the CPM employs CVFAM to assign weights to cultural values based on their relevance and fuse them into a comprehensive cultural embedding for final prediction. Through this design, CV-PM not only improves predictive accuracy but also provides interpretable insights into the cultural mechanisms underpinning crowdfunding outcomes—thereby directly responding to the theoretical and practical gaps identified in existing research.

For a clear presentation, some notions used in this paper are defined. Given a crowdfunding dataset $D = \{x_1, x_2, \ldots, x_N\}_{i=1}^{N}$, where N is the number of projects. After processing by the DCER, the culture features $F = \{x_i^1, x_i^2, \ldots, x_i^D\}_{i=1}^{N}$ are obtained, where $x_i^d \in R^{t^d}$, D is the total number of cultural values, t^d denotes the number of features in cultural value d. For each cultural value d, CNNs are utilized to map x_i^d to deep

cultural embeddings z_i^d, which captures the high-level abstraction of that cultural value. All the deep cultural embeddings $z_i^1, \ldots, z_i^D$ are later integrated through CVFAM to form a comprehensive cultural embedding z_i^c for crowdfunding prediction.

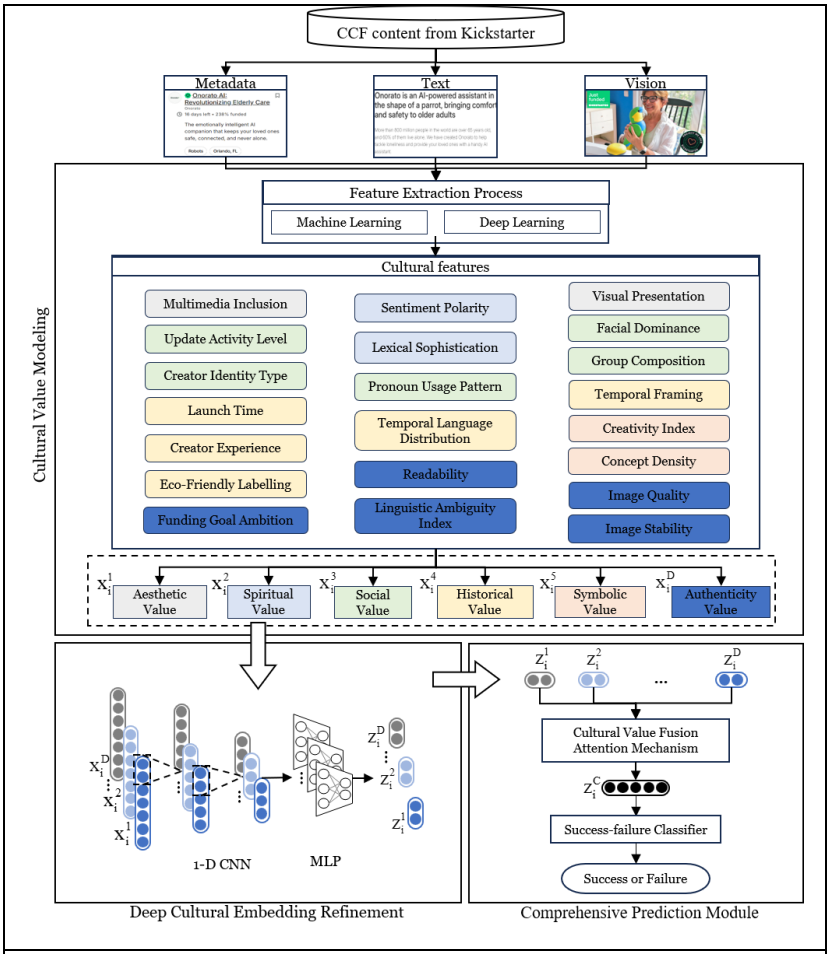


Figure 1. Cultural Value-based Prediction Model

Cultural Value Modeling

As shown in Figure 2, six cultural values are designed based on Throsby's cultural value framework, and each value is associated with a set of cultural features. Firstly, to ensure the pre-launch predictability of the method and the comprehensiveness of the features, the success determinants in the existing crowdfunding research have been sorted out to retain the pre-launch features and correspond to the connotation of Throsby's cultural value framework. Secondly, to quantify these features, metadata, text, and visual contents are subjected to a feature extraction process to obtain the cultural features that correspond to the cultural values.

Aesthetic Value

Aesthetic value refers to the properties of beauty, harmony, and other stylistic characteristics that contribute to the acknowledged quality of a cultural work (Throsby, 2001). Projects that present rich, vivid, and emotionally resonant visuals are often perceived as more professional, trustworthy, and appealing. Prior studies highlight that a higher degree of multimedia inclusion is associated with a more extroverted and affect-driven communication style, enhancing project attractiveness (Kolbe et al., 2022). Similarly, vividness indicators—such as brightness, saturation, and colorfulness—are shown to enhance perceived quality and foster emotional connections with audiences (Blanchard et al., 2023).

Based on these insights, to capture aesthetic value, we measure the multimedia inclusion from metadata, with richer visual material suggesting a more extroverted and affect-driven presentation style (Kolbe et al., 2022). Then, the brightness, saturation, and colorfulness of the images are computed using the Open source Computer Vision library (OpenCV) to assess the vividness and emotional tone of visual content (Blanchard et al., 2023). These cultural features x_i^1 collectively operationalize the aesthetic reading of crowdfunding projects i, enabling a structured evaluation of their beauty, harmony, and stylistic expression.

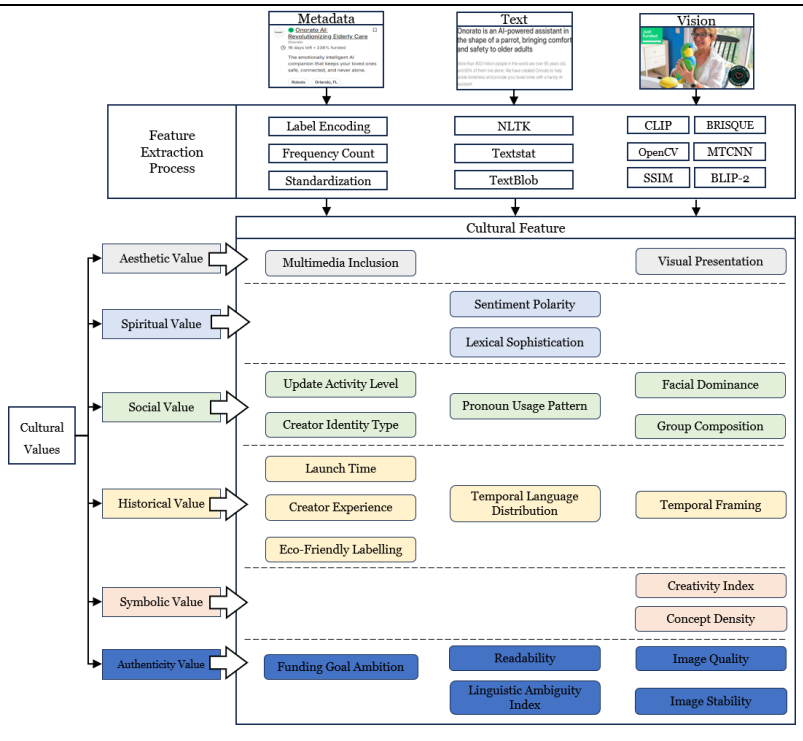


Figure 2. The Processing Procedure of Cultural Value Modeling

Spiritual Value

Spiritual value refers to the cultural significance of a work as it relates to inner human qualities such as understanding, enlightenment, and insight, beyond material or functional attributes (Throsby, 2001). In crowdfunding projects, conveying emotional depth and fostering cognitive resonance are critical for building meaningful connections with backers. Prior studies show that linguistic refinement in narratives enhances perceived thoughtfulness and credibility (Mahmood & Yeganegi, 2023), while affective sentiment significantly shapes emotional engagement and donation behavior (K. Zhao et al., 2022).

To capture spiritual value, we measure the lexical sophistication of text using the Textstat library as a proxy for the narrative's verbal refinement, where more sophisticated language suggests higher intellectual resonance (Mahmood & Yeganegi, 2023). In parallel, we calculate the sentiment polarity of text with TextBlob to assess the affective tone of the project description, where a balanced emotional expression fosters empathy and understanding (K. Zhao et al., 2022). Combined, these cultural features x_i^2 operationalize the spiritual dimension of crowdfunding project i, providing a structured representation of its emotional and cognitive appeal.

Social Value

Social value reflects the extent to which a cultural work promotes social connections, group identity, and collective belonging (Throsby, 2001). In crowdfunding projects, fostering a sense of community and shared ownership is pivotal for encouraging engagement and support. Prior research suggests that institutional affiliation, frequent project updates, and group-oriented narratives and visual design strengthen perceived

social cohesion and trust among backers (Allison et al., 2017; Courtney et al., 2017; Fang et al., 2023; Geniole & McCormick, 2015; Ghosh et al., 2019; Twenge et al., 2013).

Based on these insights, to capture social value, we extract creator type from metadata to distinguish between institutional and individual creators, where institutional affiliation often signals broader social legitimacy (Allison et al., 2017). We also measured update activity level by collecting the number of creators' past projects and the number of updates from the metadata to calculate the average update frequency, where higher update activity reflects continuous engagement and interactive communication with the community (Courtney et al., 2017). From the text, we analyzed pronoun usage patterns, where the frequent use of pronouns denoting individuals indicates the centrality of the individual, while on the contrary pronouns representing groups indicate a collective frame position (Twenge et al., 2013). Visually, we use MTCNN to extract the facial dominance and facial group composition, computed based on the size and the number of the faces in project images, where the presence of multiple faces signals group affiliation and shared agency (Fang et al., 2023; Geniole & McCormick, 2015; Ghosh et al., 2019). Combined, these cultural features x_i^3 systematically operationalize the social dimension of crowdfunding project i, capturing the ways in which creators cultivate social resonance and collective engagement.

Historical Value

Historical value pertains to a work's connection to its temporal context—how it reflects the conditions of its time and provides a sense of continuity with the past (Throsby, 2001). Building on this foundation, Throsby's later work on the sustainability of cultural capital and on culture in sustainable development emphasizes the responsibility to preserve cultural capital across both present and future generations. This extended view positions temporal significance as continuity with the past as well as stewardship toward the future (Throsby, 2005; Throsby, 2008a). In crowdfunding projects, signaling continuity, experience, and temporal orientation can enhance perceived legitimacy and future trustworthiness. Prior research suggests that indicators such as creator experience and temporal framing significantly shape backers' expectations regarding credibility, sustainability, and commitment (Hörisch, 2015; Kim & Ahn, 2019; Tran et al., 2016).

To capture historical value, we evaluate the creator's experience through the number of past projects and the total number of backers from metadata. A richer history of activity suggests a greater tolerance for innovation and a willingness to experiment (Tran et al., 2016). We further include project launch timing from metadata to contextualize the project's temporal positioning. Accordingly, we interpret eco-friendly labeling as a temporal cue for sustainability claims, expressing a commitment to long-term responsibility that aligns with the notion of historical value (Hörisch, 2015). From textual data, we tag part-of-speech categories to calculate the language distribution, including the frequency of past, present, and future verb forms, and compute the aggregate frequency of these expressions to infer the project's temporal stance using the Natural Language Toolkit (NLTK) (Loper & Bird, 2002). Visually, we assess temporal framing using Contrastive Language–Image Pre-training (CLIP), a multimodal model developed by OpenAI, which evaluates the alignment of project images with past-, present-, or future-oriented prompts. Each image receives soft scores across these categories, and the dominant class determines its temporal label (Arslan & Oz, 2017; Radford et al., 2021). Together, these cultural features x_i^4 systematically operationalize the historical dimension of crowdfunding project i, capturing how temporal cues convey continuity, experience, and alignment with evolving societal expectations.

Symbolic Value

Symbolic value reflects the meanings conveyed by a cultural work and the interpretive significance it holds for backers (Throsby, 2001). In crowdfunding projects, symbolic value is often projected through innovation, conceptual richness, and the expression of unconventional ideas, shaping how backers perceive a project's uniqueness and ideological resonance. Prior studies suggest that conceptual novelty and creative presentation significantly enhance cognitive engagement and increase project appeal (Lurie, 2004; Tran et al., 2016).

To capture symbolic value, we measure the creativity index by applying CLIP to classify each image along a spectrum of creativity, from conventional to avant-garde (Radford et al., 2021). To further enrich the measurement, we apply Bootstrapping Language–Image Pre-training 2 (BLIP-2), a vision–language model developed by Salesforce Research, to identify distinct semantic concepts in each image and compute visual

concept density (Li et al., 2023; Lurie, 2004). Together, these cultural features x_i^5 systematically operationalize the symbolic dimension of crowdfunding project i, offering a structured understanding of its conveyed meanings and conceptual depth.

Authenticity Value

Authenticity value pertains to the credibility, sincerity, and integrity perceived in cultural work, reflecting its faithfulness to its origins and its honest representation of ideas (Throsby, 2001). In crowdfunding, perceived authenticity is vital for gaining backers' trust and investment. Prior studies show that transparent communication, realistic goals, and coherent self-presentation contribute to stronger supporter engagement and project success (Mittal et al., 2012; Uddin & Chakraborty, 2022; Yin et al., 2024).

To quantify authenticity value, funding goal ambition is computed by comparing the project's funding goal to the average goal within its category, where more realistic targets are often perceived as more credible (Yin et al., 2024). From text, readability score is measured using Textstat to assess clarity and transparency in the project narrative, as clear communication reinforces perceived sincerity (Uddin & Chakraborty, 2022). Additionally, we identify the linguistic ambiguity index using the NLTK, which provides part-of-speech tagging to locate auxiliary and modal verbs that reflect linguistic uncertainty. Visually, we include two measures. Image quality, computed using BRISQUE, captures visual noise and distortion, where higher scores suggest less clarity and organization (Mittal et al., 2012). Image symmetry, calculated via Structural Similarity Index Measure (SSIM), reflects compositional balance and visual regularity, often perceived as signs of order and planning (Blanchard et al., 2023). Collectively, these cultural features x_i^6 operationalize the authenticity dimension of crowdfunding project i, enabling a structured assessment of trustworthiness, clarity, and self-coherence conveyed through multimodal content.

Following the CVM, a set of cultural features is obtained, which contains the key information used to characterize the cultural values and will be processed by the DCER and used for subsequent success prediction.

Deep Cultural Embedding Refinement

To capture high-level abstractions for each cultural value, we embed the extracted cultural features into low-dimensional representations, termed deep cultural embeddings. This step addresses two key challenges: (i) the high dimensionality and heterogeneity of multimodal features, which increase integration difficulty and computational burden; and (ii) the need to retain meaningful semantics from metadata, text, and visual contents. To address these issues, after comparison and analysis, we chose three-layer CNNs for each cultural value. To resolve these issues, we design CNNs for each cultural value. Specifically, for each cultural value $d \in \{1,2, \dots, D\}$, an independent CNN maps the corresponding features $x_i^d \in R^{t^d}$ of project i into deep cultural embeddings z_i^d. The CNNs apply three stacked 1D convolutional layers, each followed by ReLU activation, to progressively extract local semantic patterns:

$$h_{i,1}^d = ReLU\left(Conv_1\left(x_i^d\right)\right) \tag{1}$$

$$h_{i,2}^d = ReLU\left(Conv_2\left(h_{i,1}^d\right)\right) \tag{2}$$

$$h_i^d = ReLU\left(Conv_3\left(h_{i,2}^d\right)\right) \tag{3}$$

where $h_{i,1}^d$, $h_{i,2}^d$, and h_i^d denote the intermediate feature maps after each of the three convolutional layers, each followed by a $ReLU$ activation. And $Conv_1$, $Conv_2$, and $Conv_3$ denote 1D convolution operations. This progressive transformation enables the model to extract increasingly abstract semantic representations for cultural value

The final output is passed through a two-layer multilayer perceptron (MLP) to produce the deep cultural embeddings z_i^d:

$$z_i^d = W_2^d \cdot ReLU\left(W_1^d h_i^d + b_1^d\right) + b_2^d \tag{4}$$

where W_1^d, W_2^d, and b_1^d, b_2^d are the learnable weights and biases of the MLP layers for cultural value d, and z_i^d *(1, 2, …, D)* are the deep cultural embeddings.

The deep cultural embeddings retain the abstract representations of each cultural value while filtering out noise and redundancy, which are used in subsequent prediction tasks.

Comprehensive Success Prediction

While each deep cultural embedding z_i^1, z_i^2,…, z_i^D captures abstract representations of each specific cultural value, relying on them independently may lead to fragmented or biased assessments. To address this issue, we integrate these embeddings into a unified representation—referred to as the comprehensive cultural embedding z_i^c, which serves as the input for success prediction. We implement the CVFAM that dynamically assigns attention weights to each cultural value based on its relative importance to a given project. This approach not only enhances predictive performance but also improves interpretability by revealing which cultural values matter most for different types of projects.

As shown in Figure 3, we implement CVFAM to assign attention weights w_i^c to cultural value based on its importance to a given project. This mechanism computes global average pooling (GAP) and global max pooling (GMP) over deep cultural embeddings to generate summary statistics. These are concatenated and passed through a shared MLP followed by a sigmoid activation to produce attention weights.

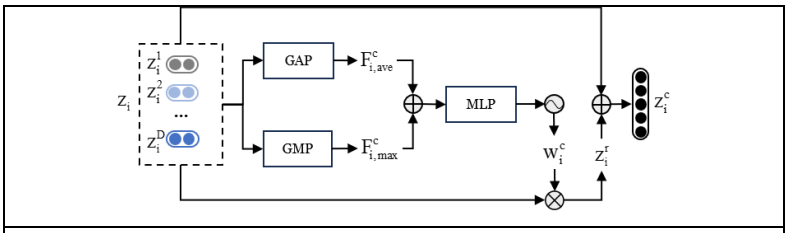


Figure 3. Cultural Value Fusion Attention Mechanism

To adaptively weigh different cultural values, we implement an attention mechanism that identifies more influential values for each project. For each cultural value $d \in \{1,2,\ldots,D\}$, let $z_i^d \in R^{m^d}$ denotes the deep cultural embedding extracted for project i, where m^d is the dimensionality of the embedding.

We first compute the GAP and GMP over each cultural value, denoted as $F_{i,ave}^c$ and $F_{i,max}^c$ respectively:

$$F_{i,ave}^c = \frac{1}{m^d}\sum_{j=1}^{m^d}\left(z_i^d(j)\right) \tag{5}$$

$$F_{i,max}^c = \max_{j=1,\ldots,m^d}\left(z_i^d(j)\right) \tag{6}$$

These two pooled vectors are concatenated along the feature axis and passed through MLP, followed by a sigmoid activation function to produce attention weights:

$$w_i^c = \sigma\left[W^c\left(F_{i,ave}^c \oplus F_{i,max}^c\right) + b^c\right] \tag{7}$$

where σ(·) is the sigmoid function, $\oplus$ denotes vector concatenation, and W^c, b^c are the learnable weights and biases of the MLP. These weights dynamically adjust the importance of each cultural value for project i. The reweighted embedding is computed via element-wise multiplication:

$$z_i^r = w_i^c(d) \cdot z_i^d \tag{8}$$

To preserve original semantics while incorporating attention emphasis, we apply a residual connection by summing the attention-enhanced embeddings with the original deep cultural embeddings—this operation is performed element-wise within each cultural value:

$$z_i^c = \left(z_i^r \oplus z_i^d\right) \tag{9}$$

Using the comprehensive cultural embedding z_i^c, we construct a prediction network to estimate the likelihood of crowdfunding success. Specifically, we employ a lightweight feedforward neural classifier that maps the embedding into a scalar probability $\hat{y}_i \in [0,1]$, representing the predicted success probability of project i:

$$\hat{y}_i = \sigma[W z_i^c + b] \tag{10}$$

where W and b are the learnable weights and biases of the classifier, and $\sigma(\cdot)$ is the sigmoid function that constrains the output to a [0,1] range. The model is trained using the binary cross-entropy loss function:

$$L_{pred} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \, log(\hat{y}_i) + (1 - y_i) \, log(1 - \hat{y}_i)] \tag{11}$$

where $y_i \in \{0,1\}$ indicates the ground truth label denoting whether project i was successfully funded. By learning from historical project outcomes, this module captures how integrated cultural values influence success likelihood, enabling interpretable success prediction in crowdfunding platforms.

In summary, we propose a CV-PM that systematically captures and integrates six dimensions of cultural value embedded in the CCF project content. By aligning feature extraction with a robust cultural theory and refining cultural features through CNNs, our method produces interpretable and task-relevant deep cultural embeddings. The fusion and prediction modules further enable nuanced modeling of value importance across projects. Overall, this framework provides a theoretically grounded and multimodal approach for predicting crowdfunding success through the lens of cultural values.

Experiment and Evaluation

Experiment Setting

Following Throsby's (2008) concentric circles model (Throsby, 2008b), we select crowdfunding projects from Kickstarter's art (Dataset 1, D1) and design (Dataset 2, D2) categories, which represent core cultural expressions and related creative industries, respectively, in CCIs. We collect projects from 2021 to 2024, resulting in a dataset of 14,539 samples (5,976 from art and 8,563 from design). Before the experiment began, we conducted two experiments and determined to select the three-layer CNNs structure in the Deep Cultural Embedding Refinement module. Firstly, a comparative experiment spanning D1 and D2 compared the performance of CNNs with that of MLP, RNN, LSTM, and Transformers. The results indicated that CNNs were the best. Secondly, the number of CNN layers on D1 and D2 was adjusted from 1 to 5. Considering both performance and cost comprehensively, 3-layer CNNs were selected.

In this study, we conduct three experiments to evaluate the effectiveness of the CV-PM. The first is a baseline comparison, where we compare our approach against traditional machine learning models including Logistic Regression (LR), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost), as well as recent multimodal deep learning models proposed, i.e., AutoGluon Multimodal (AutoGluon) (Lu et al., 2023), margin-aware distillation and modality-aware regularization (MMANet) (Wei et al., 2023), private-shared disentangled multimodal VAE (DMVAE) (Lee & Pavlovic, 2021), soft voting ensemble learning-based approach (SoftVoting) (Salur & Aydın, 2022), and PyTorch Tabular Classifier (PTC) (Hu et al., 2024). The second experiment is an ablation study designed to assess the contributions of each core module. We perform value-level ablations by excluding each of the six cultural values individually, as well as structural ablations by removing the CNN-based refinement and the CVFAM. These experiments allow us to validate the effectiveness of cultural values and the necessity of our model's key components. The third is a case study that examines the interpretability of cultural values across different domains. Specifically, we analyze the attention weights assigned by CVFAM to each cultural value within the D1 and D2, to examine the mechanism by which cultural values affect the success rate across the categories of cultural projects. All models are evaluated using three standard classification metrics—accuracy (ACC), area under the ROC curve (AUC), and F1 score—with results averaged over ten-fold cross-validation to mitigate the effect of random variance.

All models are implemented in Python 3.9 and trained using the Adam optimizer. We set the learning rate to 0.0005 and use ReLU as the activation function. The batch size is fixed at 128 for all experiments. For the Deep Cultural Embedding Refinement module, we employ a CNN architecture consisting of three 1D

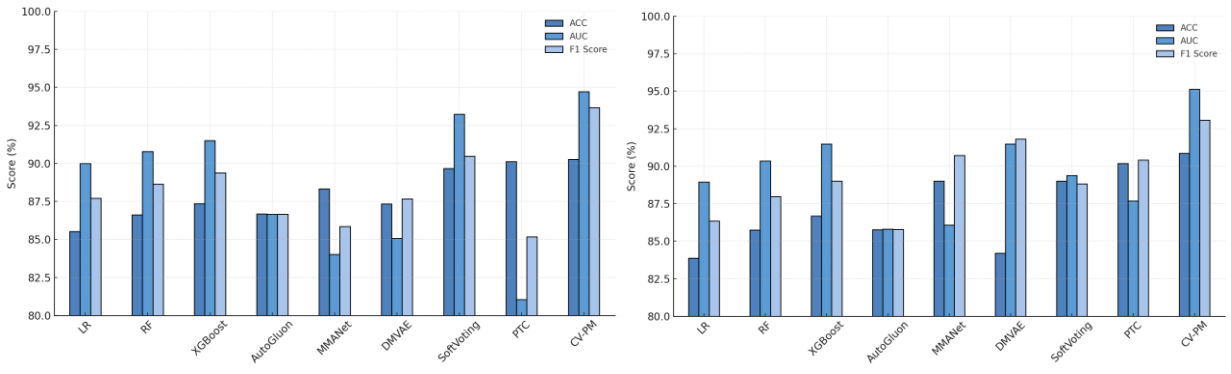
convolutional layers (Conv1d), each followed by ReLU activation. These layers are followed by a two-layer multilayer perceptron (MLP) to produce the final cultural embedding. Experiments are conducted on a server equipped with an RTX 3090 GPU (24GB), 15-core Intel® Xeon® Platinum 8362 CPU @ 2.80GHz, 60GB RAM, and a 30GB system disk.

Experiment 1: Baseline Comparison

To evaluate the effectiveness of our proposed model CV-PM, we conduct a comprehensive baseline comparison against traditional machine learning models and recent multimodal deep learning methods. As shown in Table 1 and Figure 4, CV-PM consistently outperforms all baselines across both datasets (D1 and D2) and all evaluation metrics, including ACC, AUC, and F1 Score. These results confirm the superiority of our approach in capturing the cultural values embedded in crowdfunding content.

Methods	D1			D2		
	ACC	AUC	F1 Score	ACC	AUC	F1 Score
LR	85.52	90.00	87.71	83.87	88.95	86.34
RF	86.61	90.78	88.64	85.75	90.34	87.98
XGBoost	87.36	91.50	89.38	86.67	91.48	89.01
AutoGluon	86.67	86.65	86.65	85.77	85.81	85.79
MMANet	88.33	84.01	85.85	89.01	86.08	90.72
DMVAE	87.34	85.01	87.66	84.21	91.49	91.81
SoftVoting	89.66	93.23	90.48	89.00	89.38	88.83
PTC	90.12	81.04	85.18	90.19	87.68	90.41
CV-PM	90.27	94.72	93.66	90.86	95.14	93.08

Table 1. Performance Comparison of Baseline Methods and CV-PM on ACC, AUC, and F1 Score



(a) (b)

Figure 4. Comparisons of Baseline Methods and CV-PM on Two Datasets: (a) D1, and (b) D2.

Specifically, CV-PM achieves the best performance on both datasets, demonstrating its capacity to encode expressive cultural values in art and design projects. This performance advantage is attributed to the

model's integration of interpretable cultural features and the structured fusion mechanism that dynamically weighs their importance.

Interestingly, while CV-PM delivers the best overall results, we also observe that several traditional machine learning models—LR, RF, and XGBoost—equipped with cultural value features derived from our CVM framework, outperform advanced deep learning baselines such as AutoGluon and MMANet on specific metrics. For instance, on D1, these traditional models surpass AutoGluon in both AUC and F1 scores. This suggests that the cultural value modeling itself contributes significantly to predictive performance, regardless of the model architecture, further validating the standalone strength and generalizability of the modeling of cultural values.

Experiment 2: Ablation Experiment

To evaluate the effectiveness of each component in CV-PM, we conduct an ablation study by removing modules or cultural values. Specifically, we test three types of ablations: removing the CNN-based network, the CVFAM, and each of the six cultural values individually. For this, we use two datasets: D1 for art crowdfunding (5,976 samples) and D2 for design crowdfunding (8,563 samples) from 2021 to 2024.

Methods	D1			D2		
	ACC	AUC	F1 score	ACC	AUC	F1 score
-CNNs	88.23	92.44	91.62	89.48	93.91	92.80
-CVFAM	89.28	93.74	92.66	89.66	94.11	92.95
-Aesthetic	88.09	92.74	91.53	85.67	90.48	88.01
-Spiritual	88.03	91.74	90.53	89.65	94.13	92.96
-Social	83.10	85.73	89.45	87.68	91.62	91.34
-Historical	86.13	90.47	89.59	88.50	92.94	92.12
-Symbolic	88.31	92.77	91.71	89.62	93.97	92.95
-Authenticity	88.28	92.58	91.66	89.61	93.85	92.91
CV-PM	90.27	94.72	93.66	90.86	95.14	93.08

Table 2. Results of Ablation Experiments on CV-PM

As shown in Table 2, among the modules, removing the CNNs module leads to a noticeable decline in performance (e.g., ACC drops from 90.27% to 88.23% on D1), indicating that deep cultural embeddings extracted through CNNs are crucial for cultural signal abstraction. Similarly, excluding the CVFAM results in a consistent performance drop across both datasets, demonstrating that assigning dynamic attention weights to cultural values enhances effectiveness. When excluding individual cultural values, performance drops to varying degrees depending on the dataset. In D1, removing social value leads to the largest decline in accuracy (-7.17%), indicating that project elements linked to group connection and collective identity are closely associated with success in art-related contexts. This is consistent with evidence that art backers respond strongly to social values such as community affiliation and shared identity (Bürger & Kleinert, 2021). In D2, the removal of aesthetic value results in the largest drop across multiple metrics, particularly ACC (-5.19%) and F1-score (-5.07%). This aligns with findings that aesthetic design enhances perceived quality and triggers stronger behavioral responses in product design (Candi et al., 2017). Together, these results confirm that while all cultural values matter, their relative influence varies by domain, providing insights into the underlying cultural mechanisms of project success.

Experiment 3: Case Study

To better understand how different types of cultural value contribute across domains, we further analyze the attention weights automatically learned by the CVFAM. This analysis allows us to interpret which

cultural values are emphasized most when predicting success in different project types. Specifically, we average the attention weights over all projects within each dataset—art (D1) and design (D2)—to obtain a domain-level profile of cultural value importance.

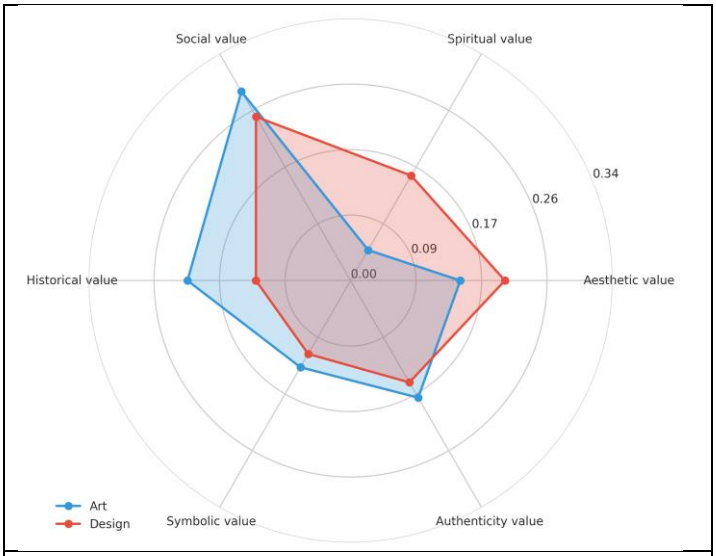


Figure 5. Radar Chart on the Importance of Cultural Values in Art and Design Projects

As shown in Figure 5, the radar chart indicates that cultural values differ in their importance across categories. First, social value has the highest weight in both art and design. Prior studies show that backers support capital-constrained cultural entrepreneurs and seek community affiliation, explaining the central role of social identification across categories (Bürger & Kleinert, 2021). Second, aesthetic value is particularly strong in design. Research in product design shows that aesthetic quality enhances perceived value and strengthens behavioral responses, with experiments confirming higher evaluations for well-executed forms. This aligns with our definition of beauty, harmony, and form, and explains the stronger contribution of aesthetic value in design (Candi et al., 2017; Shi et al., 2021). Third, historical value differs across categories, which captures a project's temporal significance, continuity with the past, and stewardship toward the future. In art crowdfunding, signals of continuity, provenance, and creator experience reduce uncertainty and strengthen legitimacy, sustainability, and trust (Hörisch, 2015; Throsby, 2005; Throsby, 2008a; Tran et al., 2016). In contrast, design crowdfunding is more strongly driven by use value, where prototyping, testing, and functionality are more diagnostic than historical narratives (Connelly et al., 2011; Darby & Karni, 1973; Nelson, 1970). Therefore, historical value contributes more in art and less in design, while social value consistently remains the strongest signal across both categories.

Conclusion, Limitations, and Future Directions

This study introduces a novel model for crowdfunding success prediction by integrating cultural theory with multimodal deep learning. Grounded in Throsby's cultural value framework, we develop the CV-PM, a theoretically guided and computationally interpretable framework that quantifies cultural values across metadata, text, and visual modalities.

Our framework comprises three key components: (i) the CVM to extract six interpretable cultural values from multimodal content; (ii) the DCER to learn high-level semantic abstractions using CNNs and MLP layers; and (iii) the CPM module incorporating CVFAM to weigh and fuse cultural signals based on their relevance to project success. Extensive experiments on 14,539 Kickstarter projects demonstrate that CV-PM consistently outperforms both traditional machine learning and recent multimodal baselines across standard classification metrics (ACC, AUC, and F1 score). Additionally, traditional models equipped with CVM-extracted features also show competitive performance, highlighting the standalone importance of the cultural features according to Throsby's cultural value framework.

Our experiments show that the influence of cultural values differs across crowdfunding categories. Ablation results confirm that each module and value contributes meaningfully to prediction, while weight visualization highlights their relative importance. Social value exerts consistent and substantial effects across categories, whereas historical value demonstrates more context-dependent influence. These patterns reveal distinct mechanisms and offer actionable guidance. In art crowdfunding, backers respond strongly to signals of community and continuity, suggesting that creators should emphasize prior work, provenance, and collective engagement, complemented by group-oriented visuals. In design crowdfunding, by contrast, support is shaped more by product-oriented cues, indicating that creators should focus on vivid visual presentations, functional prototypes, and concise language that effectively communicates benefits and feasibility. Across both domains, sustained two-way communication remains critical to fostering social ties. Building on these insights, platforms can provide practical support by creating checklists, implementing launch reviews, and refining recommendation systems that prioritize the most predictive cultural signals to improve conversion rates and pledge volumes.

Despite its contributions, CV-PM has several limitations that suggest directions for future work. First, the current framework does not incorporate backer feedback or real user signals, which may limit its ability to reflect perceived cultural salience. Integrating crowd-sourced annotations or survey data could improve the alignment between theoretical values and user interpretation. Second, CV-PM models projects based on static content at launch, overlooking how cultural signals evolve during a project. A temporal extension, such as incorporating recurrent encoders or time-aware attention, could capture the dynamic evolution of project updates and backer interactions, thereby supporting real-time insights, adaptive decision-making, and significantly enhancing real-world applicability. Third, while CV-PM currently operationalizes Throsby's cultural value framework, this fixed taxonomy may overlook emerging forms of cultural salience across platforms and contexts. To address this, we plan to explore graph-based semi-supervised learning as a complementary method, enabling the analysis to capture established cultural dimensions while flexibly adapting to newly emerging ones. Fourth, although our analysis demonstrates how cultural values shape prediction across categories and provides actionable insights for creators and platforms, it does not unpack the specific features underlying each value. Future research could examine how these features individually shape backer perception and predictive performance, as well as how multiple values interact synergistically to influence outcomes. Such analyses would deepen understanding of cultural mechanisms and yield more precise guidance for designing effective crowdfunding projects. Finally, this study does not incorporate economic evaluation indicators such as daily funding growth, overfunding rate, or backer retention. Future work will address this limitation by collecting time-series data to derive these measures and examining how cultural value features shape financial outcomes, thereby providing deeper and more actionable insights into how creators can optimize both value signaling and economic performance.

In conclusion, this work offers both theoretical and practical advancements by modeling crowdfunding through the lens of culture. It not only improves predictive performance but also empowers cultural creators and platforms with actionable insights into the values that matter.

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References

Allison, T. H., Davis, B. C., Webb, J. W., & Short, J. C. (2017). Persuasion in crowdfunding: An elaboration likelihood model of crowdfunding performance. *Journal of Business Venturing*, *32*(6), 707–725. https://doi.org/10.1016/j.jbusvent.2017.09.002

Arslan, B., & Oz, A. (2017). The effect of retro-looking products oriented consumer attitude on brand loyalty. *PressAcademia Procedia*, *4*(1), 158–170. https://doi.org/10.17261/pressacademia.2017.531

Blanchard, S. J., Noseworthy, T. J., Pancer, E., & Poole, M. (2023). Extraction of visual information to predict crowdfunding success. *Production and Operations Management*, *32*(12), 4172–4189. https://doi.org/10.1111/poms.14083

Bretschneider, U., & Leimeister, J. M. (2017). Not just an ego-trip: Exploring backers' motivation for funding in incentive-based crowdfunding. *Journal of Strategic Information Systems*, *26*(4), 246–260. https://doi.org/10.1016/j.jsis.2017.02.002

Bürger, T., & Kleinert, S. (2021). Crowdfunding cultural and commercial entrepreneurs: An empirical study on motivation in distinct backer communities. *Small Business Economics*, *57*(2), 667–683. https://doi.org/10.1007/s11187-020-00419-8

Cai, Z., Ding, H., Xu, M., & Cui, X. (2024). Multimodal dynamic graph convolutional network for crowdfunding success prediction. *Applied Soft Computing*, *154*, 111313. https://doi.org/10.1016/j.asoc.2024.111313

Candi, M., Jae, H., Makarem, S., & Mohan, M. (2017). Consumer responses to functional, aesthetic and symbolic product design in online reviews. *Journal of Business Research*, *81*, 31–39. https://doi.org/10.1016/j.jbusres.2017.08.006

Chan, H. F., Moy, N., Schaffner, M., & Torgler, B. (2021). The effects of money saliency and sustainability orientation on reward based crowdfunding success. *Journal of Business Research*, *125*, 443–455. https://doi.org/10.1016/j.jbusres.2019.07.037

Cheng, C., Tan, F., Hou, X., & Wei, Z. (2019). Success Prediction on crowdfunding with multimodal deep learning. In *28th International Joint Conference on Artificial Intelligence*, Macao, People's Republic of China. https://doi.org/10.24963/ijcai.2019/299

Cicchiello, A. F., Gallo, S., & Monferrà, S. (2023). Financing the cultural and creative industries through crowdfunding: The role of national cultural dimensions and policies. *Journal of Cultural Economics*, *47*(1), 133–175. https://doi.org/10.1007/s10824-022-09452-9

Connelly, B. L., Certo, S. T., Ireland, R. D., & Reutzel, C. R. (2011). Signaling theory: A review and assessment. *Journal of Management*, *37*(1), 39–67. https://doi.org/10.1177/0149206310388419

Courtney, C., Dutta, S., & Li, Y. (2017). Resolving information asymmetry: Signaling, endorsement, and crowdfunding success. *Entrepreneurship Theory and Practice*, *41*(2), 265–290. https://doi.org/10.1111/etap.12267

Cox, J., Nguyen, T., Thorpe, A., Ishizaka, A., Chakhar, S., & Meech, L. (2018). Being seen to care: The relationship between self-presentation and contributions to online pro-social crowdfunding campaigns. *Computers in Human Behavior*, *83*, 45–55. https://doi.org/10.1016/j.chb.2018.01.014

Darby, M. R., & Karni, E. (1973). Free competition and the optimal amount of fraud. *The Journal of Law and Economics*, *16*(1), 67–88. https://doi.org/10.1086/466756

Dematteos Guimarães, A., & Maehle, N. (2023). Evolution, trends, and narratives of cultural crowdfunding: The case of Norway. *International Journal of Cultural Policy*, *29*(7), 942–970. https://doi.org/10.1080/10286632.2022.2152446

Du, Q., Li, J., Du, Y., Wang, G. A., & Fan, W. (2021). Predicting crowdfunding project success based on backers' language preferences. *Journal of the Association for Information Science and Technology*, *72*(12), 1558–1574. https://doi.org/10.1002/asi.24530

Fang, W., Galusca, C. I., Wang, Z., Sun, Y.-H. P., Pascalis, O., & Xiao, N. G. (2023). Facial dominance augments perceived proximity: Evidence from a visual illusion. *Journal of Experimental Psychology: Human Perception and Performance*, *49*(5), 635. https://doi.org/10.1037/xhp0001102

Geniole, S. N., & McCormick, C. M. (2015). Facing our ancestors: Judgements of aggression are consistent and related to the facial width-to-height ratio in men irrespective of beards. *Evolution and Human Behavior*, *36*(4), 279–285. https://doi.org/10.1016/j.evolhumbehav.2014.12.005

Ghosh, S., Dhall, A., Sebe, N., & Gedeon, T. (2019). Predicting group cohesiveness in images. In *2019 International Joint Conference on Neural Networks (IJCNN)*. https://doi.org/10.1109/ijcnn.2019.8852184

Greenberg, J., & Mollick, E. (2017). Activist choice homophily and the crowdfunding of female founders. *Administrative Science Quarterly*, *62*(2), 341–374. https://doi.org/10.1177/0001839216678847

Greenberg, M. D., Pardo, B., Hariharan, K., & Gerber, E. (2013). Crowdfunding support tools: Predicting success & failure. In *CHI'13 Extended Abstracts on Human Factors in Computing Systems* (pp. 1815–1820). https://doi.org/10.1145/2468356.2468682

Hofstede, G. (1984). *Culture's consequences: International differences in work-related values* (Vol. 5). Sage. https://doi.org/10.1016/0377-2217(82)90018-2

Hörisch, J. (2015). Crowdfunding for environmental ventures: An empirical analysis of the influence of environmental orientation on the success of crowdfunding initiatives. *Journal of Cleaner Production*, *107*, 636–645. https://doi.org/10.1016/j.jclepro.2015.05.046

Hu, W., Yuan, Y., Zhang, Z., Nitta, A., Cao, K., Kocijan, V.,...Fey, M. (2024). Pytorch frame: A modular framework for multi-modal tabular learning. *arXiv preprint arXiv:2404.00776*. https://doi.org/10.48550/arXiv.2404.00776

Huo, H., Wang, C., Han, C. J., Yang, M., & Shang, W. L. (2024). Risk disclosure and entrepreneurial resource acquisition in crowdfunding digital platforms: Evidence from digital technology ventures. *Information Processing & Management*, *61*(3), 103655. https://doi.org/10.1016/j.ipm.2024.103655

Jones, S. (2010). Negotiating authentic objects and authentic selves: Beyond the deconstruction of authenticity. *Journal of Material Culture*, *15*(2), 181–203. https://doi.org/10.1177/1359183510364074

Kappel, T. (2008). Exante crowdfunding and the recording industry: A model for the US. *Loyola of Los Angeles Entertainment Law Review*, *29*, 375.

Kim, K., & Ahn, S. J. (2019). The moderating role of cultural background in temporal framing: Focusing on climate change awareness advertising. *Asian Journal of Communication*, *29*(4), 363–385. https://doi.org/10.1080/01292986.2019.1624793

Koh, Y., Mao-Clark, X., & DeFranco, A. (2023). Impact of socioeconomic prosperity and race on crowdfunding success. *International Journal of Contemporary Hospitality Management*, *35*(11), 4031–4050. https://doi.org/10.1108/ijchm-11-2022-1420

Kolbe, M., Mansouri, S., & Momtaz, P. P. (2022). Why do video pitches matter in crowdfunding? *Journal of Economics and Business*, *122*, 106081. https://doi.org/10.1016/j.jeconbus.2022.106081

Lee, M., & Pavlovic, V. (2021). Private-shared disentangled multimodal vae for learning of latent representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. https://doi.org/10.1109/cvprw53098.2021.00185

Li, J., Li, D., Savarese, S., & Hoi, S. (2023). Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning*. https://doi.org/10.48550/arXiv.2301.12597

Li, Y., Rakesh, V., Reddy, C. K., & Acm. (2016). Project success prediction in crowdfunding environments. In *9th Annual ACM International Conference on Web Search and Data Mining (WSDM)*, San Francisco, CA. https://doi.org/10.1145/2835776.2835791

Liu, X. F., Wang, J. H., & Zhu, J. (2022). Pricing and observational learning in crowdfunding: The moderate effect of target. *Journal of Management Science and Engineering*, *7*(1), 146–158. https://doi.org/10.1016/j.jmse.2021.12.005

Loper, E., & Bird, S. (2002, July). NLTK: The natural language toolkit. In *Proceedings of the ACL-02 Workshop on Effective Tools and Methodologies for Teaching Natural Language Processing and Computational Linguistics* Philadelphia, Pennsylvania, USA. https://doi.org/10.48550/arXiv.cs/0205028

Lu, J., Qian, Y., Zhao, S., Xi, Y., & Yang, C. (2023). Mug: A multimodal classification benchmark on game data with tabular, textual, and visual fields. *arXiv preprint arXiv:2302.02978*. https://doi.org/10.18653/v1/2023.findings-emnlp.354

Lurie, N. H. (2004). Decision making in information-rich environments: The role of information structure. *Journal of Consumer Research*, *30*(4), 473–486. https://doi.org/10.1086/380283

Mahmood, A., & Yeganegi, S. (2023). Lexical sophistication and crowdfunding outcomes. *Venture Capital*, 1–32. https://doi.org/10.1080/13691066.2023.2265565

Mason, R. (2002). Assessing values in conservation planning: Methodological issues and choices. *Assessing the Values of Cultural Heritage*, 1, 5–30.

McSweeney, J. J., McSweeney, K. T., Webb, J. W., & Devers, C. E. (2022). The right touch of pitch assertiveness: Examining entrepreneurs' gender and project category fit in crowdfunding. *Journal of Business Venturing*, *37*(4), Article 106223. https://doi.org/10.1016/j.jbusvent.2022.106223

Mittal, A., Moorthy, A. K., & Bovik, A. C. (2012). No-reference image quality assessment in the spatial domain. *IEEE Transactions on Image Processing*, *21*(12), 4695–4708. https://doi.org/10.1109/TIP.2012.2214050

Nelson, P. (1970). Information and consumer behavior. *Journal of Political Economy*, *78*(2), 311–329. https://doi.org/10.1086/259630

Porro, R., Ercole, T., Pipito, G., Vessio, G., & Loglisci, C. (2024). Pathways to success: A machine learning approach to predicting investor dynamics in equity and lending crowdfunding campaigns. *Journal of Intelligent Information Systems*. https://doi.org/10.1007/s10844-024-00883-8

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S.,...Clark, J. (2021). Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*. https://doi.org/10.48550/arXiv.2103.00020

Rykkja, A., Maehle, N., Munim, Z. H., & Shneor, R. (2020). Crowdfunding in the cultural industries. *Advances in Crowdfunding: Research and Practice*, 423–440.

Sacco, P. L. (2011). Culture 3.0: A new perspective for the EU 2014–2020 structural funds programming. *EENC Paper*.

Salur, M. U., & Aydın, İ. (2022). A soft voting ensemble learning-based approach for multimodal sentiment analysis. *Neural Computing and Applications*, *34*(21), 18391–18406. https://doi.org/10.1007/s00521-022-07451-7

Shi, A., Huo, F., & Hou, G. (2021). Effects of design aesthetics on the perceived value of a product. *Frontiers in Psychology*, *12*, 670800. https://doi.org/10.3389/fpsyg.2021.670800

Shi, S. W. (2018). Crowdfunding: Designing an effective reward structure. *International Journal of Market Research*, *60*(3), 288–303. https://doi.org/10.1177/1470785317744113

Shneor, R., Zhao, L., & Flåten, B.-T. (2020). Advances in crowdfunding: Research and practice. *Springer Nature*. https://doi.org/10.1007/978-3-030-46309-0

Solodoha, E., & Blaywais, R. (2023). Do updates matter? A comparison between rewards-based and donation-based crowdfunding platforms. *Journal of General Management*. https://doi.org/10.1177/03063070231205634

Tang, Z., Yang, Y., Li, W., Lian, D., & Duan, L. (2023). Deep Cross-Attention Network for Crowdfunding Success Prediction. *IEEE Transactions on Multimedia*, *25*, 1306–1319. https://doi.org/10.1109/tmm.2022.3141256

Throsby, C. D. (2001). *Economics and culture*. Cambridge University Press. https://doi.org/10.1017/CBO9781107590106

Throsby, C. D. (2005). *On the sustainability of cultural capital* (Vol. 510). Macquarie University, Department of Economics Sydney. https://doi.org/10.1023/A:1007543313370

Throsby, D. (2008a). Culture in sustainable development: Insights for the implementation of art. 13. *Economia Della Cultura*, *18*(3), 389–396. https://doi.org/10.1446/27881

Throsby, D. (2008b). Modelling the cultural industries. *International Journal of Cultural Policy*, *14*(3), 217–232. https://doi.org/10.1080/10286630802281772

Tran, T., Dontham, M. R., Chung, J., & Lee, K. (2016). How to succeed in crowdfunding: A long-term study in kickstarter. *arXiv preprint arXiv:1607.06839*. https://doi.org/10.48550/arXiv.1607.06839

Twenge, J. M., Campbell, W. K., & Gentile, B. (2013). Changes in pronoun use in American books and the rise of individualism, 1960–2008. *Journal of Cross-cultural Psychology*, *44*(3), 406–415. https://doi.org/10.1177/0022022112455100

Uddin, N., & Chakraborty, V. (2022). An investigation of the readability of sustainability reports. *Journal of Emerging Technologies in Accounting*, *19*(1), 69–78. https://doi.org/10.2308/JETA-18-10-01-18

Wei, F. Q., Tan, Y. L., Gao, H. B., & Zhao, H. Z. (2024). When investors meet consumers: The roles and interactions of different backers in the crowdfunding market. *Decision Sciences*, *55*(5), 474–490. https://doi.org/10.1111/deci.12612

Wei, S., Luo, C., & Luo, Y. (2023). MMANet: Margin-aware distillation and modality-aware regularization for incomplete multimodal learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. https://doi.org/10.48550/arXiv.2304.08028

Yeh, T. L., Chen, T. Y., & Lee, C. C. (2019). Investigating the funding success factors affecting reward-based crowdfunding projects. *Innovation-Organization & Management*, *21*(3), 466–486. https://doi.org/10.1080/14479338.2019.1585191

Yin, Z., Huang, G., Zhao, R., Wang, S., Shang, W.-L., Han, C., & Yang, M. (2024). Information disclosure and funding success of green crowdfunding campaigns: A study on GoFundMe. *Financial Innovation*, *10*(1), 147. https://doi.org/10.1186/s40854-024-00666-8

Zhao, K., Zhou, L., & Zhao, X. (2022). Multi-modal emotion expression and online charity crowdfunding success. *Decision Support Systems*, *163*, 113842. https://doi.org/10.1016/j.dss.2022.113842

Zhao, L., Shneor, R., & Sun, Z. (2022). Skin in the game: Self-funding and reward crowdfunding success. *Business Horizons*, *65*(1), 89–100. https://doi.org/10.1016/j.bushor.2021.09.007

Zheng, H. C., Xu, B., Wang, T., & Xu, Y. (2017). An empirical study of sponsor satisfaction in reward-based crowdfunding. *Journal of Electronic Commerce Research*, *18*(3), 269–285.

Zhong, C., Xu, W., & Du, W. (2022). Success prediction of crowdfunding campaigns with project network: A machine learning approach. *Journal of Electronic Commerce Research*, *23*(2), 99–114.